

TrES-3b

R-0104

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240409 · created 2026-07-16T03:22:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460409.8903 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.057
Transit depth (Rp/Rs)^2	2.41 +/- 1.76 [%]
Orbital Inclination (inc)	83.06 +/- 2.93
Airmass coefficient 1 (a1)	32.15 +/- 31.67
Airmass coefficient 2 (a2)	-3.0 +/- 1.67
Scatter in the residuals of the lightcurve fit is	176.12 %
Best Comparison Star	#1 - [558, 465]
Optimal Aperture	0.75
Optimal Annulus	6.75
Transit Duration (day)	0.057 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.155 $\pm$ 0.057 vs 0.16309 (deviation 0.14 σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.01 σ)
Mid-time O-C	1.4 min
Depth SNR	2.7
Residual scatter	176.12 %

Light curve

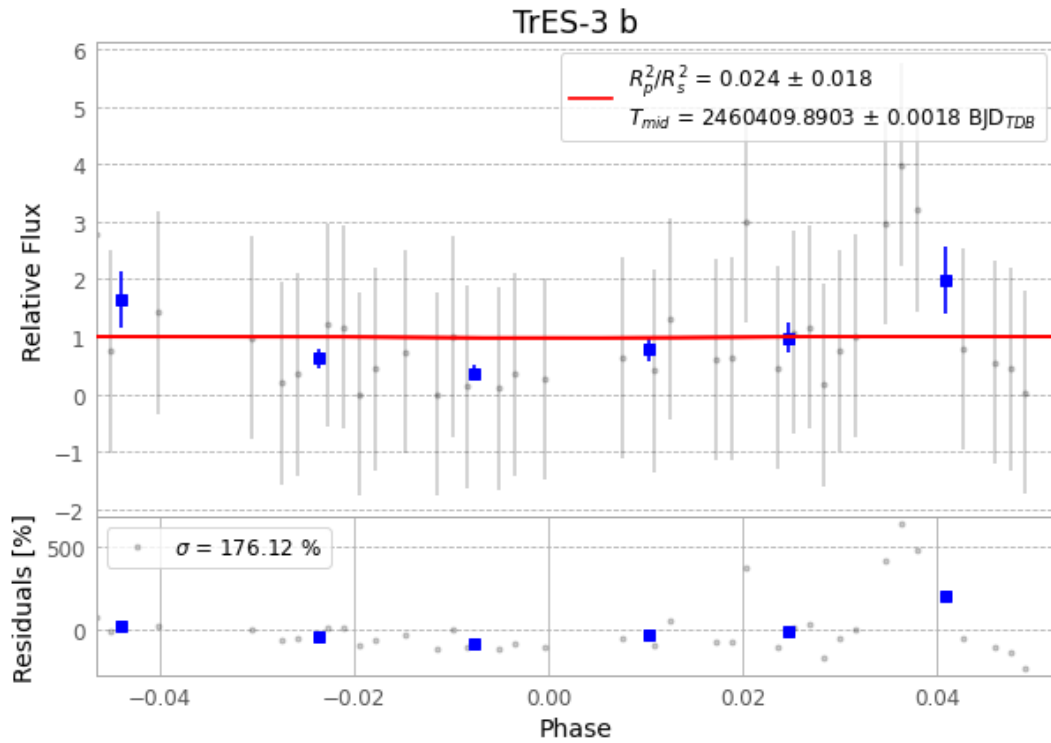


Figure 1 — FinalLightCurve_TrES-3 b_2024-04-09.png (SHA-256 ffc553170380a1b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [589, 371], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 226f51376145453e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

65 FITS frames (43 MB), TRES-3240409075115.FITS → TRES-3240409110315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240409.log (55320f913262...), FinalLightCurve_TrES-3 b_2024-04-09.png (ffc55317038...), FinalLightCurve_TrES-3 b_2024-04-09.csv (5866dd3dbf4b...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:22Z.